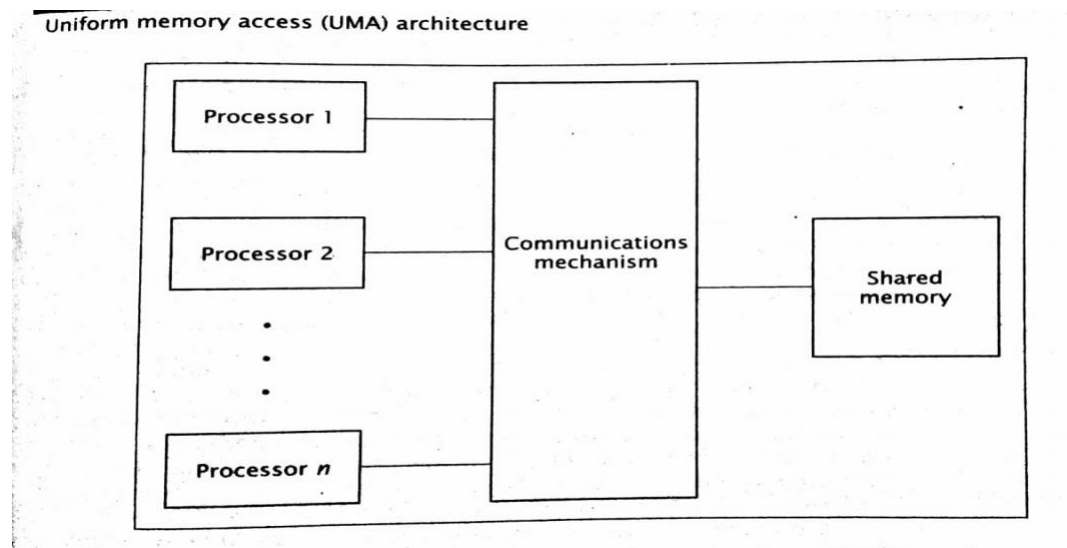


MIMD System Architectures:

- Refers to the connection of the processors w.r.t system memory.
- A **Symmetric multiprocessor**, or **SMP**, is a computer system that has two or more processors with comparable capabilities (do not have to be identical).
 - => All processors must be capable of performing the same functions; this is the symmetry of SMPs.
 - => The processors all have access to **the same I/O devices and memory modules**.

Uniform Memory Access (UMA) architecture

- One type of SMP is the Uniform **Memory Access (UMA)** architecture.
- UMA gives all CPUs equal(uniform) access to all locations in shared memory.
- They interact with shared memory via some communication mechanism.
- **Each processor may have its own cache memory**, not directly accessible by the other processors.



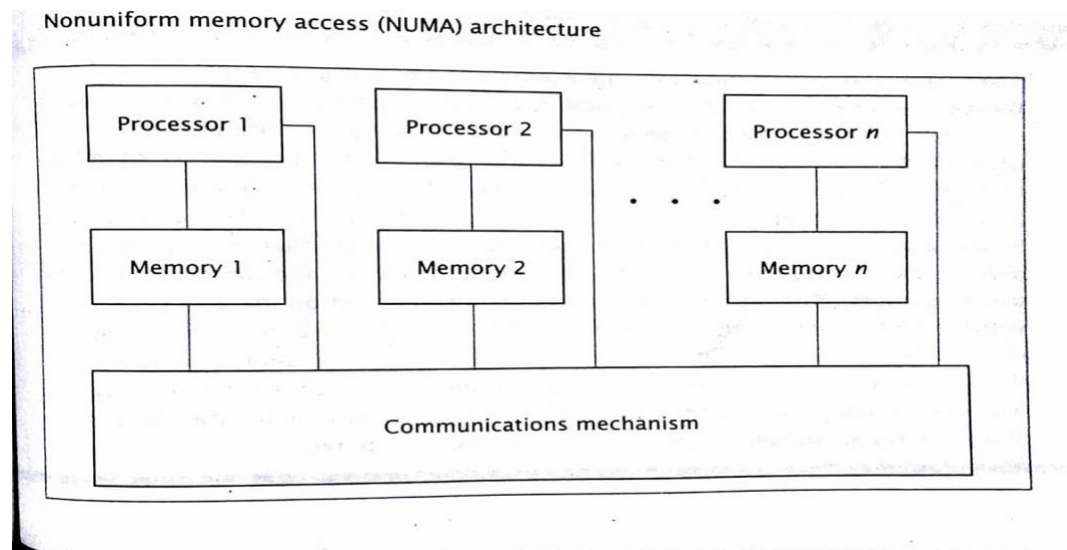
Non - Uniform Memory Access (UMA) architecture

- **Non-uniform Memory**

Access (NUMA) do not allow uniform access to all shared memory locations.

- The architecture still allows all processors to access all shared memory locations.
- However, each processor can access the memory module closest to it, its local shared memory, more quickly than the other modules; hence, the memory access times are non-uniform.

Unlike UMA machines, NUMA computers may not be SMP.



- The **cache coherence** NUMA architecture, **CC-NUMA**, is similar to the NUMA architecture , except each processor includes cache memory.
- The cache can buffer data from memory modules that are not local to the processor, thus reducing the access time of the lengthiest memory transfers.
- However, this introduces a problem when two or more caches hold the same piece of data.
- When one processor changes the value of this data, the other caches' values are invalid.
- There are several methods to maintain consistency, or **coherence**.